

Summary: The Transformer Architecture and Self-Attention

Reading group handout - Week 2

1. Motivation

Recurrent networks process tokens one at a time, which makes training slow and makes it hard to learn dependencies between distant words. The Transformer drops recurrence entirely and relies on attention to relate every position in a sequence to every other position.

2. Scaled dot-product attention

Each token is projected into a query Q , a key K and a value V . The output is $\text{softmax}(Q K^T / \sqrt{d_k}) V$. Dividing by the square root of the key dimension keeps the dot products from growing too large, which would push the softmax into tiny gradients.

3. Multi-head attention

Instead of one attention function, the model runs h heads in parallel on smaller projections and concatenates the results. Different heads learn different relations, for example syntax in one head and coreference in another.

4. Positional encoding

Attention has no notion of order, so sine and cosine signals of different frequencies are added to the input embeddings to encode each token's position.

5. Encoder and decoder

The encoder stacks self-attention and feed-forward layers with residual connections and layer normalization. The decoder adds masked self-attention so a position cannot look at future tokens, plus cross-attention over the encoder output.

6. Cost

Self-attention is quadratic in sequence length, which motivates later work on sparse and linear attention variants.

Discussion questions

- Why does the model need both residual connections and layer normalization?
- What is lost by replacing recurrence with attention?